

TruSeq® Sample Preparation Pooling Guide

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Revision History

Part #	Revision	Date	Description of Change
15042173	A	May 2013	Initial release

Introduction

This document explains Illumina® pooling guidelines when performing TruSeq® sample preparation protocols. It is a supplement to TruSeq sample preparation guides and assumes that you have reviewed your kit guides for details on performing the protocol, making sure that all of the required materials are available, and that your equipment is programmed and ready to use.



NOTE

You can download TruSeq sample preparation guides from the Illumina website at www.illumina.com. Go to the support page for the TruSeq kit you are using, then click **Documentation & Literature**.

Indexed Adapter Sequences

This section details the indexed adapter sequences.

TruSeq LT Kit Indexed Adapter Sequences

A TruSeq LT Kit or ChIP kit contains the following indexed adapter sequences.



NOTE

- The index numbering is not contiguous. There is no Index 17, 24, or 26.
- The base in parentheses () indicates the base for the seventh cycle and is not considered as part of the index sequence. The index should be recorded in the sample sheet as only six bases. For indexes 13 and above, the seventh base (in parentheses) might not be A, and this will be seen in the seventh cycle of the index read.
- For more information on the number of cycles used to sequence the index read, reference your instrument user guide.

Table 1 TruSeq LT Kit Set A Indexed Adapter Sequences for DNA Kits

Adapter	Sequence	Adapter	Sequence
AD002	CGATGT(A)	AD013	AGTCAA(C)
AD004	TGACCA(A)	AD014	AGTTCC(G)
AD005	ACAGTG(A)	AD015	ATGTCA(G)
AD006	GCCAAT(A)	AD016	CCGTCC(C)
AD007	CAGATC(A)	AD018	GTCCGC(A)
AD012	CTTGTA(A)	AD019	GTGAAA(C)

Table 2 TruSeq LT Kit Set B Indexed Adapter Sequences for DNA Kits

Adapter	Sequence	Adapter	Sequence
AD001	ATCACG(A)	AD020	GTGGCC(T)
AD003	TTAGGC(A)	AD021	GTTTCG(G)
AD008	ACTTGA(A)	AD022	CGTACG(T)
AD009	GATCAG(A)	AD023	GAGTGG(A)
AD010	TAGCTT(A)	AD025	ACTGAT(A)
AD011	GGCTAC(A)	AD027	ATTCCT(T)

Table 3 TruSeq Set A Indexed Adapter Sequences for LT RNA and ChIP Kits

Adapter	Sequence	Adapter	Sequence
AR002	CGATGT(A)	AR013	AGTCAA(C)
AR004	TGACCA(A)	AR014	AGTTCC(G)
AR005	ACAGTG(A)	AR015	ATGTCA(G)
AR006	GCCAAT(A)	AR016	CCGTCC(C)
AR007	CAGATC(A)	AR018	GTCCGC(A)
AR012	CTTGTA(A)	AR019	GTGAAA(C)

Table 4 TruSeq Set B Indexed Adapter Sequences for LT RNA and ChIP Kits

Adapter	Sequence	Adapter	Sequence
AR001	ATCACG(A)	AR020	GTGGCC(T)
AR003	TTAGGC(A)	AR021	GTTTCG(G)
AR008	ACTTGA(A)	AR022	CGTACG(T)
AR009	GATCAG(A)	AR023	GAGTGG(A)
AR010	TAGCTT(A)	AR025	ACTGAT(A)
AR011	GGCTAC(A)	AR027	ATTCCT(T)

TruSeq HT Kit Indexed Adapter Sequences

The DNA Adapter Plate (DAP) or RNA Adapter Plate (RAP) in a TruSeq HT Kit contains the following indexed adapter sequences:



NOTE

The Index recorded in the sample sheet is the full 8 bases and 8 bases are sequenced per indexed read.

Table 5 TruSeq HT Kit Indexed Adapter 1 Sequences

Adapter	Sequence	Adapter	Sequence
D701	ATTACTCG	D707	CTGAAGCT
D702	TCCGGAGA	D708	TAATGCGC
D703	CGCTCATT	D709	CGGCTATG
D704	GAGATTC	D710	TCCGCGAA
D705	ATTCAGAA	D711	TCTCGCGC
D706	GAATTCGT	D712	AGCGATAG

Table 6 TruSeq HT Kit Indexed Adapter 2 Sequences

Adapter	Sequence		Adapter	Sequence
D501	TATAGCCT		D505	AGGCGAAG
D502	ATAGAGGC		D506	TAATCTTA
D503	CCTATCCT		D507	CAGGACGT
D504	GGCTCTGA		D508	GTACTGAC

Adapter Options

Illumina provides two methods for indexing samples to perform pooled sequencing, using either DNA or RNA Adapter Index tubes or a DAP or RAP.

Adapter Tubes

A TruSeq LT Kit contains DNA or RNA Adapter Index tubes that can be used to perform pooled sequencing.

- ▶ Each tube contains a unique single 6 base index adapter on the P7 strand and contains enough reagent for 18 reactions.
- ▶ Samples prepared with these adapters can be sequenced on the MiSeq® using a 6 cycle Single Index read and any other Illumina sequencing platform using a 7 cycle Single Index read.

For more information on pooling guidelines when using adapter index tubes, see *Adapter Tube Pooling Guidelines* on page 12.

For more information on sequencing samples prepared using a TruSeq LT Kit, see your sequencing platform user guide.

Adapter Plate

A TruSeq HT Kit contains a DAP or RAP, which is a 96-well plate containing 96 uniquely indexed adapter combinations designed for manual or automated preparation of 96 uniquely indexed samples.

- ▶ Each well of the plate is single-use and the plate can undergo up to 4 freeze-thaw cycles.
- ▶ The DAP or RAP adapters provided in this plate are dual-indexed, meaning that each adapter contains two indices. These are referred to as Index 1(i7), an 8 base Index on the P7 strand, and Index 2(i5), an 8 base Index on the P5 strand.
- ▶ There are 12 Index 1 sequences (D701–D712) arrayed across the columns and 8 Index 2 sequences (D501–D508) arrayed down the rows, to generate 96 uniquely dual-indexed adapter combinations in the plate.
- ▶ If compatible, samples prepared with these adapters can be sequenced on an Illumina sequencing platform using the dual-indexed recipes for dual indexing or the 8 cycle single-indexed recipe for single indexing.

For more information on pooling guidelines when using the DAP or RAP, see *Adapter Plate Pooling Guidelines* on page 13.

For more information on sequencing samples prepared using a TruSeq HT Kit, see your sequencing platform user guide.

Table 7 Dual-Indexed Sequencing Platform Compatibility

Platform	Compatibility
MiSeq	Full compatibility
HiSeq®	• Requires TruSeq Dual Index Sequencing Primer Box, Single Read for dual-indexed sequencing on a v3 single-read flow cell.^a • Requires HCS 1.5/RTA 1.13 or later • Process with OLB 1.9.3 or later if offline base call is needed • Process with CASAVA 1.8.2 or later or HAS (any version)
Genome Analyzer™	• Requires TruSeq Dual Index Sequencing Primer Box, Single Read for dual-indexed sequencing on a single-read flow cell.^a • Requires SCS 2.10/RTA 1.13 or later • Process with OLB 1.9.4 or later if offline base call is needed • Process with CASAVA 1.8.2 or later

a. Not required for sequencing on paired-end flow cells or Rapid two lane single-read or paired-end flow cells.

Pooling Preparation with Adapter Plate

A TruSeq HT Kit contains a DAP or RAP and enables preparation of up to 96 libraries with unique dual indexes.

Figure 1 DAP or RAP Dual-Indexed Layout

	1	2	3	4	5	6	7	8	9	10	11	12
A	D701-D501	D702-D501	D703-D501	D704-D501	D705-D501	D706-D501	D707-D501	D708-D501	D709-D501	D710-D501	D711-D501	D712-D501
B	D701-D502	D702-D502	D703-D502	D704-D502	D705-D502	D706-D502	D707-D502	D708-D502	D709-D502	D710-D502	D711-D502	D712-D502
C	D701-D503	D702-D503	D703-D503	D704-D503	D705-D503	D706-D503	D707-D503	D708-D503	D709-D503	D710-D503	D711-D503	D712-D503
D	D701-D504	D702-D504	D703-D504	D704-D504	D705-D504	D706-D504	D707-D504	D708-D504	D709-D504	D710-D504	D711-D504	D712-D504
E	D701-D505	D702-D505	D703-D505	D704-D505	D705-D505	D706-D505	D707-D505	D708-D505	D709-D505	D710-D505	D711-D505	D712-D505
F	D701-D506	D702-D506	D703-D506	D704-D506	D705-D506	D706-D506	D707-D506	D708-D506	D709-D506	D710-D506	D711-D506	D712-D506
G	D701-D507	D702-D507	D703-D507	D704-D507	D705-D507	D706-D507	D707-D507	D708-D507	D709-D507	D710-D507	D711-D507	D712-D507
H	D701-D508	D702-D508	D703-D508	D704-D508	D705-D508	D706-D508	D707-D508	D708-D508	D709-D508	D710-D508	D711-D508	D712-D508

When less than the full set of 96 libraries are pooled and sequenced, it is extremely important that libraries with compatible index combinations are used in the indexed pool. Illumina strongly recommends the following planning steps before beginning library preparation:

- 1 Determine the number of libraries that will be pooled for sequencing.
- 2 Ensure that the pool contains the required index combinations, as described in *Adapter Plate Pooling Guidelines* on page 13. Select the DNA or RNA index adapters based on the same guidelines.

- 3 Use the Illumina Experiment Manager to create a sample sheet which will be used during the sequencing run. This step also identifies any incorrect index combinations, allowing re-design before library preparation starts. For more information, see the *Illumina Experiment Manager User Guide*.
- 4 Use the Illumina Experiment Manager sample plate generator to specify the layout of all sample plates in 96-well plate format for compatibility with the 96-well DAP or RAP. Arrange samples that will be pooled together in the same orientation as the indices in the DAP or RAP. For more information, see the *Illumina Experiment Manager User Guide*.

Handling Adapter Plate

The DAP or RAP is designed for use in the TruSeq Sample Prep HS protocol.

- ▶ The DAP or RAP is single-use for each well.
- ▶ Illumina recommends that the DAP or RAP does not undergo more than 4 freeze-thaw cycles.
- ▶ To maximize the use of the DAP or RAP, process more than 24 samples at a time. These samples can then be pooled in any supported configuration.

Prepare Adapter Plate

Prepare the DAP or RAP for use as follows:

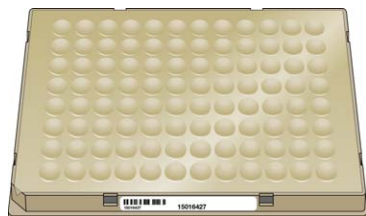
- 1 Thaw the plate for 10 minutes at room temperature on the benchtop. Visually inspect the wells to ensure that they all are completely thawed.
- 2 Remove the adapter plate tape seal.
- 3 Centrifuge the plate to 280 xg for 1 minute to collect all of the adapter to the bottom of the well.
- 4 Remove the plastic cover and save the cover if you are not processing the entire plate at once.
- 5 Apply the DAP or RAP barcode label to the DAP or RAP.
If using only part of the DAP or RAP, it may be useful to use a lab pen to mark on the foil seal the adapter wells being used. When doing so, be careful not to pierce the foil seal.

Pierce Adapter Plate Seal

Pierce the DAP or RAP foil seal as follows:

- 1 Place the DAP or RAP on the benchtop so that the part number barcode on the long side of the plate is facing you and the clipped corner is located on the lower left.

Figure 2 Correct DAP or RAP Orientation



- 2 Do one of the following:
 - If using the entire plate at once, use the bottom of a clean 96-well semi-skirted PCR plate to pierce a hole in all of the well seals simultaneously by gently but firmly pressing the clean plate over the foil seal.
 - If using only part of the plate, use the bottom of a clean eight-tube strip, with caps attached, to pierce holes in the seals of the wells that will be used for ligation. Repeat with a new, clean eight-tube strip, with caps attached, for each row or column of adapters that will be used for ligation.
- Once the foil seal has been pierced for a well, Illumina does not recommend reusing the dual-indexed adapter from that well in future sample preparations.

Pipette Adapter Plate

Pipette the adapters from the DAP or RAP into the ligation reaction as follows, while keeping the plate in the same orientation:

- 1 Using a multichannel pipette, transfer the thawed adapter from the DAP or RAP well to each well of the sample plate.
- 2 Change pipette tips between wells of the DAP or RAP. This is critical to avoid cross-contamination between wells.
- 3 Aspirate each dual-indexed adapter by column or row depending on the adapters being used.
- 4 Discard the tips after pipetting into the ligation reaction.

Adapter Plate Storage

If not all adapter wells are used in a single experiment (< 96 samples), the plate can be stored for future use of unused wells as follows:

- 1 Wipe the foil seal covering unused wells with a sterile 70% Ethanol wipe.
- 2 Allow the foil seal to dry.
- 3 Put the plastic cover that came with the DAP or RAP back on the plate.
- 4 Store at -15° to -25°C.



NOTE

Do not reseal the plate with a disposable seal. This will rip the original foil seal when the disposable seal is removed for future uses.

Pooling Guidelines

Illumina uses a green laser to sequence G/T and a red laser to sequence A/C. At each cycle at least one of two nucleotides for each color channel needs to be read to ensure proper image registration. It is important to maintain color balance for each base of the index read being sequenced, otherwise index read sequencing could fail due to registration failure. Follow these low plex pooling guidelines, depending on the TruSeq Sample Prep kit you are using.

Adapter Tube Pooling Guidelines

When using the index adapter tubes from a TruSeq LT Kit or ChIP Kit, follow these pooling guidelines for single-indexed sequencing. TruSeq LT Kit Sets A and B, each contain 12 unique index adapter tubes. When designing low-plexity index pools for single-indexed sequencing, always use at least two unique and compatible barcodes for each index sequenced. The following table describes possible pooling strategies for 2–4 samples generated with the adapter index tubes in each set.

- ▶ For 5–11plex pools, use 4-plex options with any other available adapters
- ▶ Not all color-balanced pools are listed. Check the color balance of such user-designed pools using the Illumina Experiment Manager's sample sheet generator.

Table 8 Single-Indexed Pooling Strategies for 2–4 Samples with DNA Kits

Plexity	Option	Set A Only	Set B Only
2	1	AD006 and AD012	Not recommended
	2	AD005 and AD019	
3	1	AD002 and AD007 and AD019	AD001 and AD010 and AD020
	2	AD005 and AD006 and AD015	AD003 and AD009 and AD025
	3	2-plex options with any other adapter	AD008 and AD011 and AD022
4	1	AD005 and AD006 and AD012 and AD019	AD001 and AD008 and AD010 and AD011
	2	AD002 and AD004 and AD007 and AD016	AD003 and AD009 and AD022 and AD027
	3	3-plex options with any other adapter	3-plex options with any other adapter

Table 9 Single-Indexed Pooling Strategies for 2–4 Samples with RNA and ChIP Kits

Plexity	Option	Set A Only	Set B Only
2	1	AR006 and AR012	Not recommended
	2	AR005 and AR019	
3	1	AR002 and AR007 and AR019	AR001 and AR010 and AR020
	2	AR005 and AR006 and AR015	AR003 and AR009 and AR025
	3	2-plex options with any other adapter	AR008 and AR011 and AR022
4	1	AR005 and AR006 and AR012 and AR019	AR001 and AR008 and AR010 and AR011
	2	AR002 and AR004 and AR007 and AR016	AR003 and AR009 and AR022 and AR027
	3	3-plex options with any other adapter	3-plex options with any other adapter

For more information on the Single-Indexed Sequencing workflow, see the Illumina HiSeq, HiScan®, Genome Analyzer, and MiSeq user guides.

Adapter Plate Pooling Guidelines

When using a DAP or RAP from a TruSeq HT Kit, follow these pooling guidelines. In addition, please review *Pooling Preparation with Adapter Plate* on page 9 and *Handling Adapter Plate* on page 10.

Single-Indexed Sequencing

Follow the single-indexed sequencing workflow when pooling 12 or fewer samples. When designing low plexity index pools, always use at least two unique and compatible barcodes for each index sequenced. The following figures illustrate possible pooling strategies for 2-12 samples generated with the DAP or RAP.

- ▶ Color balanced pools are shaded light gray with green wells.
- ▶ For 5-plex pools, dark gray wells are not used for pooled sequencing. They are available for individual sequencing.
- ▶ For 7-11plex pools, combine any of the 2-6plex pools.
- ▶ Not all color-balanced pools are illustrated. Check the color balance of such user-designed pools using the Illumina Experiment Manager's sample sheet generator.

For more information on the single-indexed sequencing workflow, see the Illumina HiSeq, HiScan, Genome Analyzer, and MiSeq user guides.

Figure 3 Single-Indexed-2-plex

D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●

Figure 4 Single-Indexed-3-plex

D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●

Figure 5 Single-Indexed-4-plex

D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●

Figure 6 Single-Indexed-5-plex

D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●

Figure 7 Single-Indexed-6-plex

D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●
●	●	●	●	●	●	●	●	●	●	●	●

Figure 8 Single-Indexed-12-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
	●	●	●	●	●	●	●	●	●	●	●	●
	●	●	●	●	●	●	●	●	●	●	●	●
	●	●	●	●	●	●	●	●	●	●	●	●
	●	●	●	●	●	●	●	●	●	●	●	●
	●	●	●	●	●	●	●	●	●	●	●	●
	●	●	●	●	●	●	●	●	●	●	●	●
	●	●	●	●	●	●	●	●	●	●	●	●
	●	●	●	●	●	●	●	●	●	●	●	●

Dual-Indexed Sequencing

Follow the dual-indexed sequencing workflow when pooling more than 12 samples into one pool. When designing the low-plexity index pools, always use at least two unique and compatible barcodes for each index sequenced. The following figures illustrate possible pooling strategies for 2–16 samples generated with the DAP or RAP.

- ▶ Color balanced pools are shaded light gray with green wells. The 2-plex pools are diagonal and shaded in light or dark gray with green wells.
- ▶ Odd numbered pools display dark gray wells that are not used for pooled sequencing. They are available for individual sequencing.
- ▶ Not all color-balanced pools are illustrated. Check the color balance of such user-designed pools using the Illumina Experiment Manager's sample sheet generator.

For more information on the dual-indexed sequencing workflow, see the Illumina HiSeq, HiScan, Genome Analyzer, and MiSeq user guides.

Figure 9 Dual-Indexed-2-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 10 Dual-Indexed-3-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 11 Dual-Indexed-4-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 12 Dual-Indexed-5-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 13 Dual-Indexed-6-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 14 Dual-Indexed-7-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 15 Dual-Indexed-8-plex, Option 1

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 16 Dual-Indexed-8-plex, Option 2

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 17 Dual-Indexed-12-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Figure 18 Dual-Indexed-16-plex

	D701	D702	D703	D704	D705	D706	D707	D708	D709	D710	D711	D712
D501	●	●	●	●	●	●	●	●	●	●	●	●
D502	●	●	●	●	●	●	●	●	●	●	●	●
D503	●	●	●	●	●	●	●	●	●	●	●	●
D504	●	●	●	●	●	●	●	●	●	●	●	●
D505	●	●	●	●	●	●	●	●	●	●	●	●
D506	●	●	●	●	●	●	●	●	●	●	●	●
D507	●	●	●	●	●	●	●	●	●	●	●	●
D508	●	●	●	●	●	●	●	●	●	●	●	●

Technical Assistance

For technical assistance, contact Illumina Technical Support.

Table 10 Illumina General Contact Information

Illumina Website	www.illumina.com
Email	techsupport@illumina.com

Table 11 Illumina Customer Support Telephone Numbers

Region	Contact Number	Region	Contact Number
North America	1.800.809.4566	Italy	800.874909
Austria	0800.296575	Netherlands	0800.0223859
Belgium	0800.81102	Norway	800.16836
Denmark	80882346	Spain	900.812168
Finland	0800.918363	Sweden	020790181
France	0800.911850	Switzerland	0800.563118
Germany	0800.180.8994	United Kingdom	0800.917.0041
Ireland	1.800.812949	Other countries	+44.1799.534000

MSDSs

Material safety data sheets (MSDSs) are available on the Illumina website at www.illumina.com/msds.

Product Documentation

Product documentation in PDF is available for download from the Illumina website. Go to www.illumina.com/support, select a product, then click **Documentation & Literature**.



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